

诚信应考,考试作弊将带来严重后果!

华南理工大学期末考试

《20241128 Calculus Review Derivative Application》试卷

注意事项: 1. 考前请将密封线内填写清楚;

2. 所有答案请直接答在试卷上(或答题纸上);

3. 考试形式: 闭卷;

4. 本试卷共 8 大题, 满分 100 分, 考试时间 120 分钟。

题号	1, 2	3, 4	5, 6	7, 8	9, 10	总分
得分						
评卷人						

1. Answer the questions (26):

(1) If $f(x) = nx(1-x)^n$, $x \in [0, 1]$ $n \in N$, find $\text{Max} f(x)$ and $\lim_{n \rightarrow \infty} \text{Max} f(x)$

(2) Find the maximum and minimum $f(x) = \int_0^{x^2} (2-t)e^{-t} dt$, $x \in (-\infty, \infty)$,

(3) If $f(x)$ continue on $[a, b]$ and $\varphi(x) = \left| x - \frac{a+b}{2} \right| - \left| f(x) - \frac{a+b}{2} \right|$

Show there exists $\xi \in [a, b]$ s.t. $f(\xi) = \xi$ if $\varphi(a) \geq 0$, $\varphi(b) \leq 0$

And $\left| \int_a^b f(x) dx - \frac{b^2 - a^2}{2} \right| \leq \frac{(b-a)^2}{4}$, if $\varphi(x) \geq 0$, $x \in [a, b]$

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(4) If $f(x)$ is differentiable on $[0, 1]$ and $f(1) = 1$

Show there exists $\xi \in [0, 1]$ s.t. $\xi f'(\xi) + 2f(\xi) = 3\xi$

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(5) $f(x) > 0$ continue, $x \in [a, b]$. Show there exists $\xi \in [a, b]$, $\int_a^\xi f(t) dt = \int_\xi^b \frac{1}{f(t)} dt$

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(6) If $f''(x) > 0$, then $f\left(\frac{x+y}{2}\right) < \frac{f(x) + f(y)}{2}$, ($a > 0$)

(7) $f''(x) > 0$, then $f\left(\frac{x_1 + \dots + x_n}{n}\right) < \frac{f(x_1) + \dots + f(x_n)}{n}$

(8) Find the $f'_-(0)$ and $f'_+(0)$, $\tan x > x + \frac{x^3}{3}$, $\left(0 < x < \frac{\pi}{2}\right)$

(9) If $f''(x) > 0$, then $f(x) > f(x_0) + f'(x_0)(x - x_0)$